

Riyad Cornu

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EDUCATION

Toulouse INP – ENSEEIHT (N7)

Toulouse, France

Engineering degree, Systems & Networks Architecture (ASR) track

2025 – 2028 (expected)

- Core curriculum: computer architecture, networks & telecommunications, language translation / compilation, concurrent & distributed systems, formal methods (Why3, Coq).

CPGE MPI, Lycée Paul Valéry

Paris, France

Preparatory classes — Mathematics, Physics, Computer Science

2024 – 2025

CPGE MP2I-MPI, Lycée Hoche

Versailles, France

Preparatory classes — Mathematics, Physics, Computer Science

2022 – 2024

EXPERIENCE

Basic Fit

Vélizy, France

Front Desk Agent

06/2024 – 09/2024 and 06/2025 – 09/2025

- Managed customer relations, membership sales, and after-sales support.
- Resolved conflicts and ensured proper functioning of the club facilities.

Auchan Drive

Vélizy, France

Multi-skilled Team Member

02/2022 – 07/2022

- Prepared customer orders and managed stock and inventory levels.

PROJECTS

Neural Network from Scratch

2025

- Built a neural network from scratch in CUDA, with no ML framework, including hand-written GPU kernels for every matrix operation.
- Implemented backpropagation and the Adam optimizer; reached 96% accuracy on MNIST and 89% on EMNIST.

Unix Mini-shell

2026

- Unix shell in C: multi-command pipelines, I/O redirection (</>), background execution (&), and builtins (cd, exit, rep).
- Process management via fork/execvp and pipes (pipe/dup2); asynchronous synchronization through a SIGCHLD handler (sigaction) and sigsuspend, with no busy-waiting or zombie processes.

Memory Allocator (malloc)

2026

- Reimplemented malloc/free/calloc/realloc in C directly on top of Linux primitives (mmap, sysconf), with no dependency on libc.
- Linked-list block allocator with splitting and reuse of freed regions; validated with a unit test suite (memory alignment, multiple allocations, calloc, realloc, edge cases).

CHIP-8 Emulator

2026

- Wrote a CHIP-8 interpreter in C, implementing the full instruction set (opcode decoding, timers, stack, registers).
- Used SDL2 for display rendering and keyboard input; validated against the IBM logo, Corax+, and BC_test ROMs.

CERTIFICATIONS

Project Management MOOC (GdP27)

March 2026

- Project Management MOOC (Rémi Bachelet, Centrale Lille), 27th session — specialization modules "MRP" (Problem-Solving Methodology) and "GPAS" (Agile Project Management with Scrum) — mooc.gestiondeprojet.pm

SKILLS & INTERESTS

Technical: C, Rust, Ada, CUDA (systems & low-level); Python, MATLAB (applied/scientific computing); Why3, Coq (formal methods)

Languages: French (native), English (B2/C1)

Interests: Strength training & powerlifting, reading